Ecole Supérieure d'Informatique et du Numérique
COLLEGE OF ENGINEERING & ARCHITECTURE

TD 06: Fundamental Security
Instructor: Prof. Anass Sebbar
Level: Engineering - 3rd Year

Tutorial: Firewall Rules and Network Security Analysis

Objectives:

The purpose of this tutorial is to understand and configure firewall rules in a stateless packet forwarding firewall. You will analyze how different network security mechanisms work and evaluate the effectiveness of firewall policies in preventing unauthorized access. Additionally, the lab will introduce key cybersecurity concepts, including malware classification, encryption techniques, and authentication protocols.
By the end of this tutorial, you will be able to:

- Define and apply firewall rules for different network services.
- Analyze the behavior of firewalls when handling different types of network traffic.
- Identify security vulnerabilities and understand their implications in network defense.
- Familiarize yourself with fundamental cybersecurity concepts, such as malware types, encryption algorithms, and authentication mechanisms.

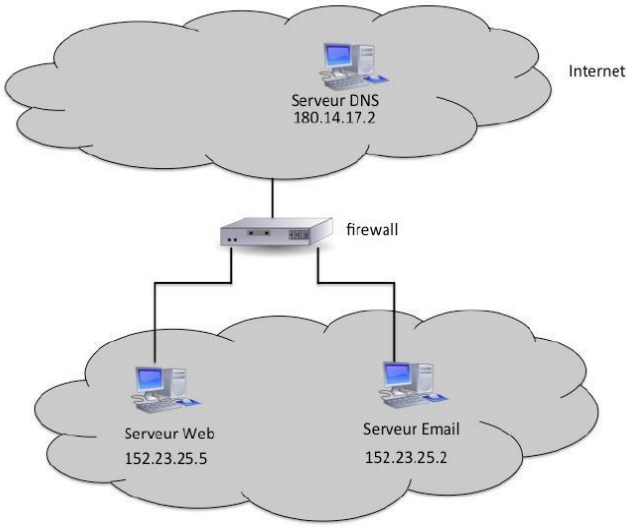
Instructions

- The tutorial report must be submitted on the Moodle platform within one week of the lab session, adhering to the specified deadline.
- The tutorial must be performed individually during the session, but the report should be submitted in groups of up to two students.

Legal and Ethical Precautions

- Configuring and testing firewalls in real-world environments without authorization is **strictly prohibited**. Ensure that all activities are conducted in a controlled and authorized setting.
- Any form of penetration testing or security analysis must be **performed only on systems for which you have explicit permission**. Unauthorized network intrusion is illegal and unethical.
- This lab is designed for **educational purposes only**. Do not attempt to apply these techniques outside a supervised academic environment.

Règles Firewall

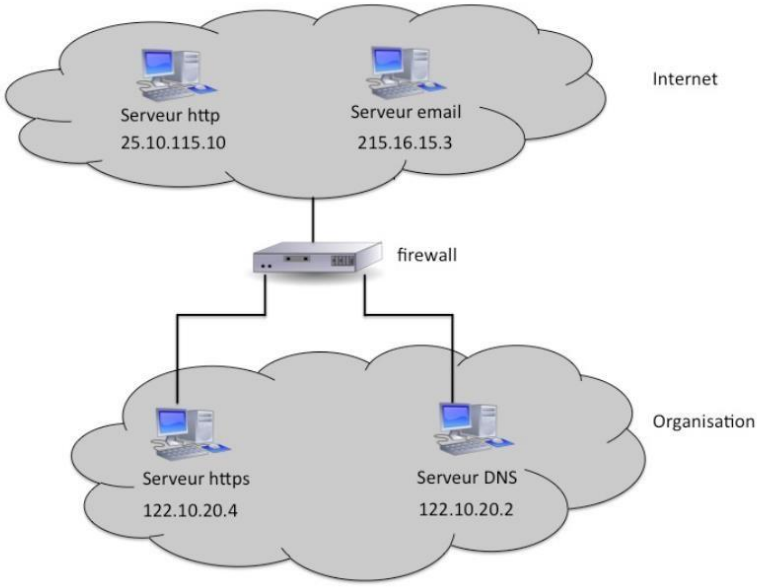


1- Specify the firewall rules for this company. Let's assume it's a stateless packet forwarding firewall. (DNS port is 53, http port is 80, mail sever port is 25)

IP source	Port source	IP destination	Port destination	protocol

2- Let's assume that the firewall is confugured according to your response in qst 1.
   a- Can the packets with the following characteristics go through the firewall (IP source=180.14.17.2, port source=53, IP destination= 153.23.25.5, port destination=80, protocol=80) ?
   b- What if the packet contains a virus?

Configuration d'un firewall



We have the organization above

1- Complete the firewall rules. (http port : 80email server port : 25,https port : 443 and DNS port : 53)

IP source	Port source	IP destination	Port destination	protocol

2- The email server send an email with :

IP source : 122.10.20.4
Port source :443
IP destination : 122.10.20.2
Port destination : 53
Can the packet go through the firewall ? Explain.

Terminology

Fill in the table with the following terms:
WPA, Diffie-Hellman, Trojan Horse (trojan), Worm (wom), Backdoor, AES, SSL, MAC, Wireshark, SHA1

No	Term	Description
1		End-to-end security protocol
2		Protocol aimed at securing Wi-Fi communication
3		Malware that allows an attacker to connect remotely to a machine
4		Algorithm used to authenticate messages
5		Malware that spreads by sending itself from one machine to another
6		Hashing algorithm
7		Asymmetric encryption algorithm
8		Software that allows network sniffing
9		Malware that appears to be useful
10		Symmetric encryption algorithm

True/False

Answer true or false and justify your response.

1. A stateful firewall can protect against more attacks than a stateless firewall.
2. All malware necessarily exploits a software vulnerability such as buffer overflow.
3. It is possible to write a program in such a way that it does not have any buffer overflow vulnerability.
4. By updating to the latest virus signatures, one can be sure that their computer cannot be infected by viruses.
5. It is possible to find malware that can be classified as both a worm and a backdoor.